

Arpit Verma

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CAREER OBJECTIVE

Computer Science (AI & ML) undergraduate with experience in MERN stack development, data structures & algorithms, and machine learning fundamentals. Built and deployed real-world projects using the MERN stack, MongoDB Atlas, and modern web technologies. Passionate about software engineering, problem-solving, and developing scalable applications.

SKILLS

Programming Languages: Python, C, C++

Frontend: HTML5, CSS3, JavaScript, React.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Databases: MySQL, Firebase, MongoDB

Libraries & Tools: Axios, Chart.js, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Core Concepts: Data Structures & Algorithms, OOPs, Operating Systems, DBMS, API Integration

PROJECTS

Smart Queue Management System | GitHub | Live Demo

- Developed and deployed a full-stack Smart Queue Management System using Node.js, Express.js, MongoDB Atlas, and **Socket.io**.
- Implemented **JWT** authentication, **QR code**-based token tracking, and real-time queue status updates.
- Designed a multi-counter queue allocation system with live monitoring of waiting, serving, and completed tokens.
- Built an analytics dashboard and deployed the application on **Render** with **MongoDB Atlas** integration.

Smart Water Automation and Dosing System | GitHub | Live Demo

- Developed a **patent-published IoT-based automation system** integrating **ESP32** and **Python** for real-time monitoring and control.
- Designed a **MERN web-based dashboard** enabling real-time monitoring, control, and **data visualisation**.
- Implemented automated decision-making logic for chemical dosing using real-time sensor inputs.
- Achieved **patent publication in Indian Patent Journal (Feb 2026)**.

CryptoTracker: Real-time Cryptocurrency Tracker | GitHub | Live Demo

- Developed a high-performance **React.js** web application to track **500+ cryptocurrencies** with real-time market data.
- Integrated real-time cryptocurrency market data APIs to display live prices, trends, and market statistics.
- Integrated **CoinGecko API** using **Axios** to fetch dynamic data, including price, market cap, and 24-hour fluctuations.
- Implemented **global state management** using **Redux Toolkit**, ensuring efficient data flow and minimising unnecessary re-renders.
- Designed a **responsive, mobile-first UI** using **Tailwind CSS**, enhancing accessibility across devices.
- Built **interactive data visualisations** using **Chart.js** for historical trend analysis (24h, 7d, 30d).

EDUCATION

Pranveer Singh Institute of Technology, Kanpur

Bachelor of Technology in Computer Science (AI&ML).

Coursework: Data Structures and Algorithms (DSA), AI & ML Foundations, Computer Architecture, SQL, Web Development.

Patent App No: 202511131860 An IoT-based automatic water quality monitoring and dosing system.

Technical Profile: Building full-stack applications using the MERN stack.

ACHIEVEMENTS & CERTIFICATIONS

- **Problem Solving:** Solved **200+ DSA problems** on LeetCode.
- **Technical Workshop:** Attended workshop at **IIT Kanpur** on emerging technologies and applications.
- **Certifications:** Infosys Springboard HTML, CSS, JavaScript, Python | Learning **Full Stack (Node.js, Express)**
- **PROTECH 2K25:** Presented **Travel In-Depth** project at PSIT.